Open Science and Data Management Plan

FINESST proposal companion document (2-page limit). Anonymized section — no names or institutional identifiers. Verify final question list against ROSES F.5 §5.1.1.3 before submission.

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1. Data and software the project will use

All input data are free, public, and archived by their originating programs; nothing is purchased or restricted.

Input	Archive / host	Access
2001 airborne lidar DEM (NASA ATM, 2 m)	USGS ScienceBase	public
2014–15 airborne lidar DEM + point cloud (NCALM, 1 m)	OpenTopography	public
REMA satellite DEM v2 (2 m mosaic, 2021–23)	Polar Geospatial Center / AWS Open Data	public
Stream discharge (21 gauges), meteorology, glacier mass balance, soil climate	MCM-LTER via the Environmental Data Initiative (EDI)	public
Stream-channel outlines (labels)	EDI (`knb-lter-mcm` scope)	public
ERA5 reanalysis	Copernicus Climate Data Store	public (free account)
AMPS Antarctic mesoscale model output	NCAR GDEX	public

One dataset in this domain is *not* public: the prior dissertation's hand-digitized training tiles, held by their author. If shared, they will be used under the author's terms and will not be redistributed; any project products derived with them will be released in forms that do not reproduce the tiles themselves. If not shared, the project digitizes its own training set — which, unlike the original, will be published (§2).

All processing software is open source: Python (numpy/scipy, rasterio, geopandas, PyTorch), PDAL, GDAL/OGR, and WhiteboxTools. No proprietary software is required to reproduce any result.

2. Data and software the project will produce

Product	Format / standard	Where released
Elevation-change (DoD) rasters, all epoch pairs	Cloud-Optimized GeoTIFF, explicit CRS (EPSG:3294), nodata and units in metadata	Zenodo (DOI per release)
Per-stream erosion/deposition rate tables	CSV with data dictionary	Zenodo; offered to EDI for co-listing with the LTER stream data they derive from
Per-pixel calibrated detection-threshold layers (O3)	Cloud-Optimized GeoTIFF	Zenodo
Stream-channel outlines + process-classified geomorphic-change maps (O2)	GeoPackage (OGC) / Cloud-Optimized GeoTIFF	Zenodo
Training labels (if self-digitized)	GeoPackage + tile rasters	Zenodo — closing the label-access gap this field currently has
Trained model weights	PyTorch checkpoint + ONNX export, with training configuration	Zenodo
Processing code (full pipeline: fetch → derivatives → detection → change → attribution)	Python, version-controlled public repository, OSI-approved license (Apache-2.0)	Public repository, archived to Zenodo (DOI) at each release

Product	Format / standard	Where released
Driver-attribution model outputs (O1)	CSV + fitted-model objects with documentation	Zenodo, with the O1 publication

Documentation standard. Every released raster/vector product carries: coordinate reference system (EPSG code), acquisition epochs and processing date, method summary, uncertainty estimate (NMAD/LOD95 or the O3 per-pixel layer), and a pointer to the exact code version (commit/DOI) that produced it.

3. When and where products are shared

- Code: developed in the open in a public repository from award start; released

versions archived with DOIs.

- Data products: released no later than the publication of the paper they support,

and at yearly milestones regardless of publication status (Yr 1: driver tables + Taylor Valley rates; Yr 2: cross-valley detection products; Yr 3: uncertainty layers + synthesis products).

- Publications: submitted to open-access venues or archived as accepted manuscripts

in a public repository, consistent with SMD open-access policy.

4. Licenses

Data products: CC0 or CC-BY. Software: Apache-2.0. Third-party inputs remain under their original terms and are cited, not re-hosted, except where their license permits convenience copies.

5. Protections and exceptions

The project uses no human-subjects data, no export-controlled data, and no commercially restricted data. The single access-limited item (the author-held label tiles) is handled as described in §1: use with permission, no redistribution, and a published replacement set if the project digitizes its own.

6. Roles and oversight

The FI is responsible for day-to-day data management, repository hygiene, and product releases; the PI reviews each public release. Working data are stored on institutionally backed storage with an off-machine mirror; the public archives (Zenodo, EDI) provide long-term preservation beyond the award period.

7. Reproducibility commitment

Every number and figure in project publications will regenerate from public inputs via the released code: a single documented command sequence per product. Heavy intermediate rasters are treated as regenerable and are not archived; the scripts and small tables/figures that define them are.